Education

Labour Economics, ECON207 — UG-II, 2026: Semester IV

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Sections Covered

Table 1: Borjas, *Labor Economics* 9E — Chapter 6

§	Topic
6-1	Education in the Labor Market: Stylized Facts
6-2	Present Value
6-3	The Schooling Model
6-4	Education and Earnings
6-5	Estimating the Rate of Return to School
6-6	Policy: School Construction in Indonesia
6-7	Policy: The Education Production Function
6-8	Do Workers Maximize Lifetime Earnings?
6-9	Signaling

What Is Human Capital?

- People bring a unique set of abilities and acquired skills into the labour market — this stock of productive capacity is called **human capital**.
- Workers add to their human capital throughout their lives, especially through **formal education** and **on-the-job experience**.
- Unlike physical capital, human capital is **embodied in the worker** — it cannot be separated from the person who carries it.
- The human capital framework treats schooling as an **investment decision**: workers weigh current costs against future returns.

How Is Education Linked to Labour Market Outcomes?

- **Labour force participation**: In 2021, college graduates had LFP rates near 85%, compared to ~60% for high school dropouts (Borjas, Table 6-2).
- **Unemployment**: The unemployment rate for college graduates was 3.6%, versus ~10% for those without a high school diploma.

- **Earnings**: College graduates earned about $89,000 annually; high school dropouts earned about $30,000 — a ratio of nearly 5:1.
- These three stylized facts are **consistent across gender and race/ethnicity**, though levels differ.

Different Perspectives on Human Capital

- **Becker (1964)**: Human capital is directly useful in production — education increases productivity across tasks.
- **Schultz / Nelson-Phelps (1966)**: Human capital enhances **adaptability** — educated workers adjust faster to technological change.
- **Bowles & Gintis (1976)**: Education reflects the ability to work in organisations and follow orders — a "correspondence principle."
- **Spence (1973)**: Observable human capital serves as a **signal** of innate ability rather than an independently productive trait.

What Is Present Value?

- Investments in education require **spending now** for **returns later** — we need a way to compare dollars across time.
- The **PV** of y dollars received t periods from now, discounted at rate r:

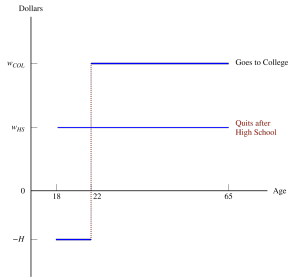
$$PV = \frac{y}{(1+r)^t}$$

- r is the **rate of discount** — it reflects both the market interest rate and the individual's **time preference** (impatience).
- Higher r means **less patient**: future earnings are worth less today, making education less attractive.

What Are the Earnings Streams Facing a High School Graduate?

- A high school graduate at age 18 faces a **choice**:
- **Option 1**: Enter the labour market immediately and earn w_{HS} from age 18 until retirement.
- **Option 2**: Attend college for 4 years, paying direct costs H per year and forgoing w_{HS} in earnings, then earn w_{COL} from age 22 until retirement.
- The **total cost of college** = direct costs (H) + **opportunity cost** (w_{HS} foregone for 4 years).
- She should attend college if the **present value of the college earnings stream exceeds the present value of the high school earnings stream**.

What Are the Earnings Streams Facing a High School Graduate? (Figure)



☒ How Is the College Decision Formalised?

- **Present value of HS stream** (Borjas, Eq. 6-4):

$$PV_{HS} = \sum_{t=0}^{T} \frac{w_{HS}}{(1+r)^t}$$

- **Present value of college stream** (Borjas, Eq. 6-5):

$$PV_{COL} = \sum_{t=0}^{3} \frac{-H}{(1+r)^t} + \sum_{t=4}^{T} \frac{w_{COL}}{(1+r)^t}$$

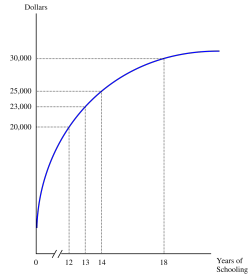
- **Decision rule** (Borjas, Eq. 6-6): Go to college iff $PV_{COL} > PV_{HS}$.
- Equivalently: go to college iff the **internal rate of return** exceeds r.

What Is the Wage-Schooling Locus?

- The **wage-schooling locus** shows the salary a particular worker would earn at each level of schooling.
- **Three key properties**:
- It is **upward sloping**: more schooling ☒ higher wages.

- The **slope** at any point gives the **percentage increase in earnings** from one more year of school.
- It is **concave**: there are **diminishing returns** to schooling — the 13th year of school raises wages less than the 10th.
- Concavity means the **marginal rate of return to schooling (MRR)** is declining.

What Is the Wage-Schooling Locus? (Figure)



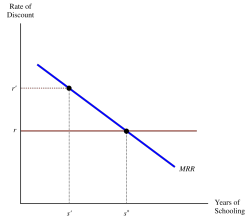
How Does a Worker Choose How Much Schooling to Get?

- The **marginal rate of return (MRR)** schedule plots the percentage return to each additional year of school — it slopes **downward** due to concavity of the wage-schooling locus.
- A worker maximises the present value of lifetime earnings by going to school until:

$$\boxed{MRR = r}$$

- The **stopping rule** (Borjas, Eq. 6-9): acquire schooling up to the point where the marginal return equals the rate of discount.
- A worker with discount rate r chooses s^* years of schooling.

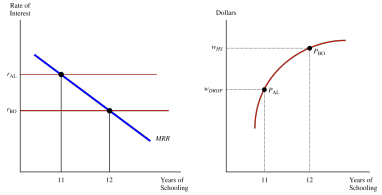
How Does a Worker Choose How Much Schooling to Get? (Figure)



☒ Why Do Workers with Different Discount Rates Get Different Amounts of Education?

- All workers face the **same** wage-schooling locus (same ability), but differ in **patience**.
- Worker **Al** has a low discount rate r_{Al} ☒ invests in more schooling ☒ earns more.
- Worker **Bo** has a high discount rate r_{Bo} ☒ invests in less schooling ☒ earns less.
- The observed earnings differential between Al and Bo **traces out the wage-schooling locus** — the slope of the observed earnings-schooling line equals the true rate of return to school.
- **Key insight**: When workers differ **only** in discount rates, the observed correlation between education and earnings **correctly identifies** the causal return.

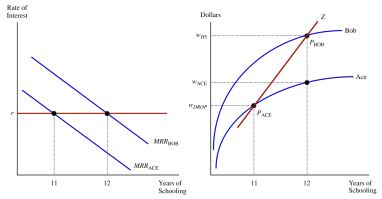
Workers with Different Discount Rates (Figure)



What Happens When Workers Differ in Ability?

- Now suppose workers differ in both **discount rates** and **ability**. More able workers face a **higher** wage-schooling locus — they are more productive at every level of schooling.
- Worker **Ace** (high ability): higher locus, gets more schooling, earns more. Worker **Bob** (low ability): lower locus, gets less schooling, earns less.
- The observed wage gap ΔW between Ace and Bob reflects **both** the return to additional schooling **and** the ability premium.
- The observed line connecting Ace and Bob is **steeper** than either worker's true wage-schooling locus — the OLS estimate is **upward biased**.

Workers with Different Abilities (Figure)



What Is Ability Bias?

- **Ability bias**: OLS overstates the return to schooling because **unobserved ability** drives both education and wages.
- In a regression of log wages on years of schooling:

$$\log w_i = a + b \cdot s_i + \text{other variables}$$

- The coefficient b captures the true return **plus** the ability premium.
- If we could control for ability A_i:

$$\log w_i = a + b \cdot s_i + \alpha \cdot A_i + \text{other variables}$$

  then b would measure the true return. But ability is typically **unobserved**.

☒ How Big Is the Ability Bias in Practice?

- The **Mincer earnings equation** (Borjas, Eq. 6-10):

$$\log w_i = a + b \cdot s_i + c_1 \cdot x_i + c_2 \cdot x_i^2 + \varepsilon_i$$

  where x_i = experience. Typical OLS estimate: $\hat{b} \approx 0.09$ (9%/year).
- **Omitted variable bias formula**: $\hat{b}_{OLS} = b_{true} + \alpha \cdot \delta$, where α = ability effect on wages and δ = correlation of ability with schooling.
- Both $\alpha > 0$ and $\delta > 0$ ☒ OLS is **upward biased**.
- Solutions: **twin studies** (difference out genetic ability) and **instrumental variables** (find exogenous variation in schooling).

How Do Twin Studies Address Ability Bias?

- **Identical twins** share the same genetic endowment ($A_i = A_j$), so differencing within twin pairs eliminates the ability term:

$$\Delta \log w_j = b \cdot \Delta s_j + \text{other variables}$$

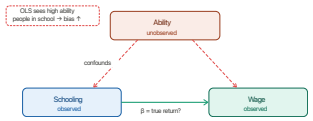
- This removes the **genetic component** of ability bias (Borjas, Eq. 6-12).
- Twin study estimates range from **3% to 15%** annual return per year of schooling — a wide range.
- **Problems**: measurement error in self-reported schooling (biases b ☒ zero); twins may differ in non-genetic traits (motivation, health).

How Do Instrumental Variables Address Ability Bias?

- **Instrumental variables (IV)**: find an exogenous source of variation in schooling that is **uncorrelated with ability**.

- **Quarter of birth** (Angrist & Krueger 1991): compulsory schooling + school-entry cutoffs ☒ Q1-born drop out earlier. IV estimate: −7.5%/year.
- **French riots of 1968**: easier admission ☒ exogenous education increase for one cohort. IV estimate: −14%/year.
- **Compulsory schooling laws**: changes in minimum school-leaving age provide exogenous variation across cohorts.

The Problem: Ability Confounds the Schooling–Wage Link



The IV Solution: Use Only the "Clean" Variation

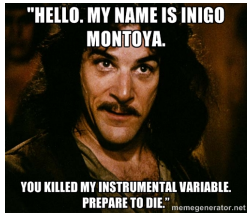
The Logic of IV: Strip Away Ability, Keep Only the Nudge

- **The problem**: $\log(w) = \beta \cdot s + \gamma \cdot A + \varepsilon$ — but A is **unobserved**
- **The instrument** Z must satisfy:
- **Relevance**: Z shifts s
- **Exclusion**: Z affects w **only through** s — no direct path, no correlation with A
- **2SLS mechanics**:

- Stage 1: $s = \pi Z + v$ ⊠ predicted $\hat{s}$ contains only **ability-free** variation
- Stage 2: $\log(w) = \beta\,\hat{s} + \varepsilon$ ⊠ clean estimate

Classic Instruments and the Exclusion Restriction

- **Card (1995)**: proximity to a college — geography nudges enrollment, not ability
- **Angrist & Krueger (1991)**: quarter of birth — compulsory schooling cutoffs assign education by birth timing
- **The hard part**: **"uncorrelated with ability" is usually untestable** — you can't directly verify the exclusion restriction
- IV estimates are only as credible as the **institutional story** about why Z is truly exogenous to unobservables



The Best Instruments for the Mincerian Wage Equation

Controlling for unobserved ability A in $\log w = a + b \cdot s + \gamma A + \varepsilon$

Instrument (Z)	Study	Why It Works	OLS	IV	Country
Quarter of birth	Angrist & Krueger (1991) *QJE*	Compulsory schooling + school-entry cutoffs ⊠ Q1-born drop out earlier; birth timing ≈ random w.r.t. ability	7.1%	7.6–8.9%	US

Instrument (Z)	Study	Why It Works	OLS	IV	Country
College proximity	Card (1995)	Growing up near a 4-year college lowers attendance cost; geography ≠ ability. LATE for credit-constrained compliers.	7.3%	**13.2%**	US
Compulsory schooling laws	Harmon & Walker (1995) *AER*	UK raised min. leaving age (14⊠15 in 1947, 15⊠16 in 1973); exogenous cohort variation	6.1%	**15.2%**	UK
School construction (INPRES)	Duflo (2001) *AER*	61,000 schools built in high-poverty districts; DiD across districts × birth cohorts	8.1%	6.8–10.6%	Indonesia
French riots of 1968	Maurin & McNally (2008) *JLE*	May 1968 protests ⊠ *baccalauréat* exam cancellation ⊠ easier university admission for one cohort	6–7%	**~14%**	France
Identical twins	Ashenfelter & Krueger (1994) *AER*; Ashenfelter & Rouse (1998) *QJE*	Within-twin differencing eliminates genetic ability; co-twin's report as IV corrects measurement error	6.7–8.4%	**12–16%**	US

- **Pattern** (Card 2001, *Econometrica*): IV estimates typically **≥ OLS** — downward measurement-error bias often exceeds upward ability bias. IV also captures **LATE** for compliers with high marginal returns.

⊠ What Did the Indonesia School Construction Experiment Show?

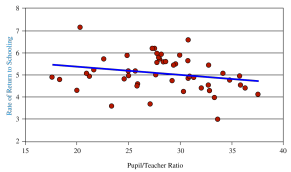
- In 1973, Indonesia launched **INPRES**: a massive school construction program that built over 61,000 schools in high-poverty regions.
- **Identification strategy** (Duflo 2001): Difference-in-differences comparing (1) high-construction vs. low-construction districts × (2) young cohort (exposed) vs. old cohort (not exposed).
- Results: Additional construction increased schooling by **0.11 years** and earnings by **3%**.
- **Implied IV return**: $0.03/0.11 \approx 27\%$ (raw), or about **10% after controls** — one of the cleanest developing-country estimates.

Does School Quality Affect the Return to Education?

- Not all years of schooling are equal — **school quality** matters.

- Card & Krueger (1992): States with lower **pupil-teacher ratios** (more resources per student) had **higher rates of return** to schooling.
- The **Maimonides Rule** (Angrist & Lavy 1999): In Israel, class sizes are capped at 40. When enrollment crosses a multiple of 40, a new class is added, creating a sharp drop in class size. This natural experiment shows **smaller classes raise test scores**.
- **Tennessee STAR**: Randomised small classes raised test scores. **Dale & Krueger (2002)**: after controlling for *which schools admitted you*, elite college premium ≈ **zero**.

Does School Quality Affect Returns? (Figure)



Can We Test Whether Workers Made the Right Choice?

- The schooling model predicts that each worker selects the level of education that **maximises the present value of lifetime earnings**.
- To test this, we would need to observe the **counterfactual**: what would the college graduate have earned had she *not* gone to college?
- But once a choice is made, the **alternative earnings stream is never observed**.
- Naively comparing college and HS earnings is **contaminated by selection bias**: workers **self-select** into the option that best suits their **comparative advantage**.

⊠ The Willie and Wendy Example

- **Willie** (blue-collar advantage): HS = \$150K, College = \$140K ⊠ chooses HS.
- **Wendy** (white-collar advantage): HS = \$100K, College = \$175K ⊠ chooses college.
- Naive comparison: Wendy \$175K vs. Willie \$150K ⊠ "college premium" = \$25K.
- But **Willie's true return to college** = $140K - 150K = -\$10K$ (negative!). The observed premium is **meaningless** for any given individual.

Signaling in the Animal Kingdom

Economists didn't invent signaling — evolution did.

Why Does the Peacock Work but the Pufferfish Doesn't?

	Peacock	Fiddler crab	Pufferfish
Signal	Giant tail	Regrown big claw	Inflate body
Costly?	Very (predation risk)	Cheap (hollow muscle)	Cheap
Honest?	Yes — only fit males survive	No — looks strong, actually weak	Partially — smart predators ignore it
Economics parallel	Degree from a tough university	Fake degree / diploma mill	Credential inflation (everyone has an MBA)

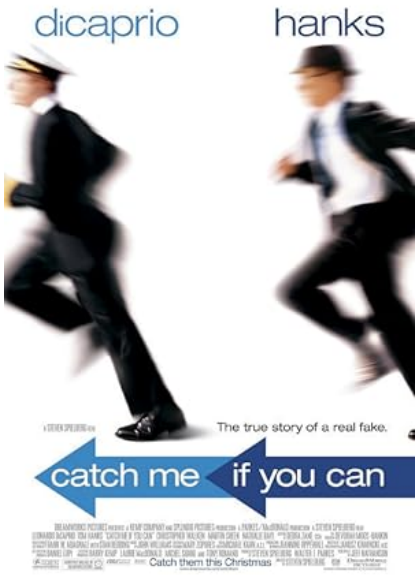
- **The pattern**: A signal is credible only when it is **too costly for low-quality types to fake**.
- When the cost collapses ⊠ everyone signals ⊠ the signal becomes **noise**.
- This is exactly what **Michael Spence (1973)** formalised for the labour market…

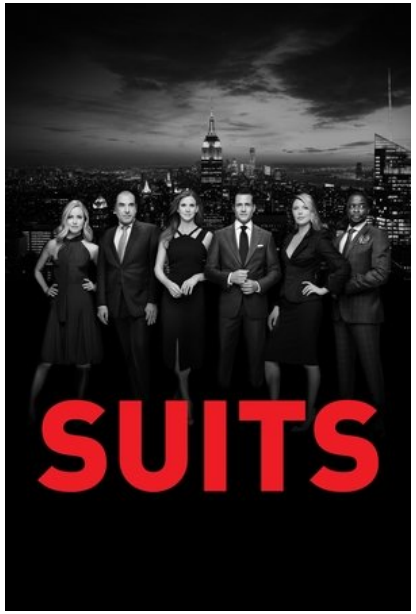
What Is the Signaling Model?

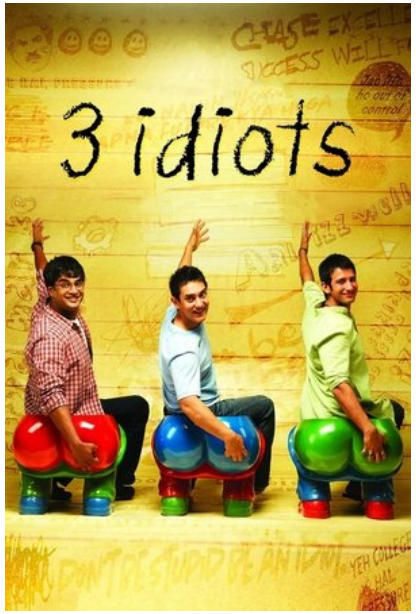
- **Key assumption**: Education does **not** increase a worker's productivity — it merely reveals the worker's **innate ability** to employers.

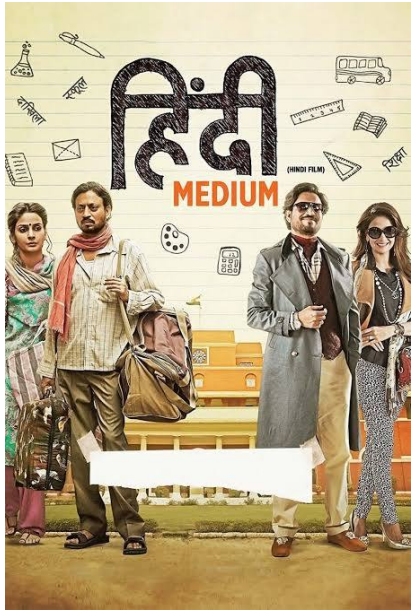
- Workers **know** their own ability, but employers **cannot observe it** directly ⊠ **asymmetric information**. Education serves as a **signal** — just like the **peacock's tail**.
- For a signal to work, it must be **more costly for low-ability workers** to acquire — otherwise everyone would signal (the **pufferfish problem**).
- This gives rise to two possible equilibria: **pooling** and **separating**.

Signaling at the Movies: The Credential Is Everything









- **Catch Me** — fake credentials, zero human capital
- **Suits** — real skills, no degree

- **3 Idiots** — Chatur = signal; Rancho = human capital
- **Hindi Medium** — school name signals, not education quality

When Human Capital Wins Back

(a) Good Will Hunting (1997)

(a) Legally Blonde (2001)

(a) 12th Fail (2023)

- **Good Will Hunting** — raw ability wasted until mentorship unlocks it
- **Legally Blonde** — starts as signal, ends as genuine skill
- **12th Fail** — fails 12th but clears UPSC; exam reveals ability

What Is a Pooling Equilibrium?

- Suppose employers **cannot distinguish** high-ability and low-ability workers.

Share of workforce	True productivity
q (low-productivity)	\$200,000
$(1-q)$ (high-productivity)	\$300,000

- $$\text{Average salary} = 300{,}000 - 100{,}000 \cdot q$$

- All workers are paid the **same pooled wage** regardless of ability — high-ability workers are **underpaid**, low-ability workers are **overpaid**.
- High-ability workers have an **incentive to signal** their type by acquiring education, even if education doesn't increase productivity.

What Is a Separating Equilibrium?

- Firms announce: **workers with** $\geq y$ years of college get \$300K; **workers with** $< y$ years get \$200K.
- For this to be a **separating equilibrium**, two conditions must hold:
- **Low-productivity workers must not want to get** y years of college:
- Therefore, the low-productivity worker will not attend college if:

$$\$200{,}000 > \$300{,}000 - (\$25{,}001 \times y)$$

- Solving: $25{,}001y > 100{,}000$ ⊠ $y > 3.999$ years.
- **Rule of thumb**: if the firm requires more than 4 years of college, no low-productivity worker will ever attend — it is just too expensive.

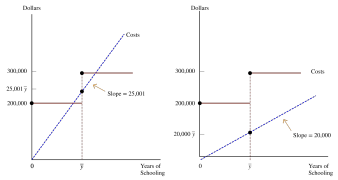
What Is a Separating Equilibrium?

- **High-productivity workers must want to get** y years of college:
- The net salary without college is $\$200{,}000$. Cost for high-ability: $\$20{,}000 \times y$.
- Therefore, high-productivity workers get y years whenever:

$$\$200{,}000 < \$300{,}000 - (\$20{,}000 \times y)$$

- Solving: $20{,}000y < 100{,}000$ ⊠ $y < 5$ years.
- **Separating equilibrium exists** for $4 < y < 5$ years (Borjas, Eq. 6-23).

Separating Equilibrium (Figure)



(a) Low-Productivity Workers (b) High-Productivity Workers

Which Threshold Survives Competition?

- Any threshold in the range $4 < y < 5$ generates a valid separating equilibrium — firms could pick 4.333, 4.5, or 4.997 years and both types would sort themselves accordingly.
- But **high-productivity workers prefer the lowest threshold**: all firms pay the same \$300K, so workers gain nothing from extra years beyond the minimum required.
- **Competitive pressure** eliminates higher thresholds — firms offering 4.333 years lose talent to firms requiring only 4 years, since both pay \$300K.
- The **competitive equilibrium** converges to the **smallest feasible threshold**: a **4-year college diploma** becomes the separating device.

⊠ Why Must Signaling Be Differentially Costly?

- For education to be a **credible signal**, it must satisfy the **single-crossing property**:

$$\frac{\partial^2 C}{\partial s \, \partial A} < 0$$

  where $C(s, A)$ = cost of s years for ability A. More able workers find it **cheaper** to acquire education.
- If all workers faced the **same cost**, low-ability workers would mimic high-ability ones ⊠ **pooling**, not separating.
- The **Spence condition**: the marginal cost of signaling must be strictly decreasing in ability.
- **Real-world manifestation**: high-ability students find coursework easier, graduate faster, suffer less disutility from studying ⊠ education is genuinely cheaper for them in time, effort, and psychic cost.

What Are the Implications of the Signaling Model?

- For schooling to act as a signal, schooling must be **more costly for low-ability workers** than for high-ability workers.
- **Private returns**: Even if education only signals, individual workers still earn more by getting educated ⊠ the private return to schooling is **positive**.
- **Social returns**: If education is *purely* a signal, the social return could be **zero** — but sorting itself has value, so the social return is likely positive even in the signaling model.
- **Reality**: Education is probably **both** — partly human capital (raises productivity) and partly signal (sorts workers). The debate is about the **relative magnitudes**.

Review — Human Capital Basics

- Education is an **investment**: workers weigh current costs (tuition + foregone earnings) against future returns.
- More education ⊠ higher LFP, lower unemployment, higher earnings — monotonically, across all groups.

- Four views of why: Becker (productivity), Schultz (adaptability), Bowles-Gintis (compliance), Spence (signal of innate ability).

Review — The Schooling Model

- Total cost of college = direct costs (H) + opportunity cost (foregone w_{HS} for 4 years).
- Stopping rule: invest in schooling until **MRR = r** (marginal return equals discount rate).
- The wage-schooling locus is concave — diminishing returns mean each extra year adds less.
- If workers differ only in patience, the observed education-earnings slope correctly identifies the causal return.

Review — Ability Bias

- **Ability bias**: unobserved ability drives both schooling and wages, so OLS overstates returns (upward bias).
- Line Z connecting high-ability and low-ability workers is steeper than either worker's true wage-schooling locus.
- Mincer equation: $\log w = a + b \cdot s + c_1 x + c_2 x^2$ — OLS gives $\hat{b} \approx 9\%$/year, but this includes the ability premium.

Review — Estimating Returns

- Twin studies difference out genetic ability; measurement error in self-reported schooling biases the estimate back down.
- IV uses exogenous shocks to schooling (quarter of birth, college proximity, compulsory laws, school construction).
- IV estimates are typically **≥ OLS** — measurement-error bias (downward) often exceeds ability bias (upward).
- School quality matters (smaller classes ⊠ higher scores), but elite college premium ≈ zero after controlling for selection (Dale-Krueger).

Review — Selection & Comparative Advantage

- The **counterfactual** (what you would have earned on the other path) is never observed — fundamental problem of causal inference.
- Workers self-select based on comparative advantage: Willie chooses HS because his true college return is negative (−\$10K).
- Naive comparisons of college vs. HS earnings confound selection with returns — the observed "premium" is meaningless for any given individual.

Review — Signaling

- In the signaling model, education **reveals** ability rather than raising productivity — the peacock's tail, not a training programme.
- Separating equilibrium requires the **single-crossing property**: schooling must be cheaper for high-ability types so low types cannot mimic.
- Separating range: $4 < y < 5$ years; competition drives firms to the minimum threshold (the 4-year degree).

- Private returns are positive even under pure signaling; social returns may be near zero (credential arms race).
- Reality: education is probably both human capital and signal — the debate is about relative magnitudes.